

LIMBERT, Antarctica

WMO: 892570

Lat: 75.914S

Lon: 59.264W

Elev: 40

StdP: 100.85

Time Zone: -3.00 (W03)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-49.4	-47.2	-54.0	0.0	-49.3	-51.8	0.0	-47.2	26.2	-22.7	25.7	-23.2	2.9	220	1.335

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
	DB Range	DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	6.2	0.2	-0.9	-0.9	-1.8	-1.9	-2.6	-1.4	-0.7	-2.3	-1.7	-3.2	-2.6	3.8	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
-2.2	3.2	-1.0	-3.2	2.9	-1.9	-4.1	2.7	-2.7	7.4	-0.6	5.8	-1.6	4.3	-2.5	6.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.0	12.6	10.7	-51.9	4.0	1.3	2.6	-52.8	5.8	-53.6	7.3	-54.4	8.8	-55.3	10.6		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-22.8	-6.6	-13.9	-23.2	-28.6	-30.4	-31.9	-33.6	-32.7	-28.9	-24.2	-13.3	-6.3		
	DBStd	11.58	3.36	5.36	6.84	6.41	7.26	7.37	7.34	7.57	7.06	6.39	4.87	3.43		
	HDD10.0	11984	515	669	1029	1158	1251	1257	1351	1323	1168	1059	698	506		
	HDD18.3	15026	773	902	1287	1408	1509	1507	1609	1581	1418	1318	948	764		
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	4.4	4.0	4.8	4.7	4.5	4.0	3.6	4.0	4.3	4.4	5.1	4.8	4.2		
(15) Precipitation	PrecAvg	417	22	23	28	27	37	37	43	51	57	40	31	21		
	PrecMax	589	43	45	59	52	99	75	123	82	99	89	73	48		
	PrecMin	275	3	4	5	7	7	7	10	11	22	6	3	3		
	PrecStd	78	10	10	12	14	18	19	21	17	16	19	15	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.1	-0.8	-3.9	-10.1	-8.2	-8.9	-8.5	-6.9	-10.4	-5.2	-0.9	2.3		
		MCWB	2.1	-1.7	-4.3	-10.4	-8.2	-9.2	-8.5	-6.9	-10.8	-6.0	-2.2	0.2		
	2%	DB	0.7	-2.6	-7.0	-14.0	-12.5	-12.8	-14.9	-12.5	-12.7	-9.3	-3.3	0.1		
		MCWB	-0.4	-3.1	-7.3	-14.2	-12.9	-13.2	-15.2	-12.8	-13.0	-9.6	-4.0	-1.2		
	5%	DB	-0.7	-3.9	-9.8	-16.9	-15.8	-17.3	-18.2	-17.6	-15.6	-11.9	-4.6	-0.8		
		MCWB	-1.5	-4.2	-10.0	-17.2	-16.0	-17.5	-18.5	-17.8	-16.0	-12.3	-5.2	-1.8		
	10%	DB	-1.9	-5.5	-13.1	-19.4	-19.8	-21.4	-22.5	-21.3	-18.5	-14.4	-5.8	-1.7		
		MCWB	-2.5	-5.9	-13.5	-19.6	-20.0	-21.7	-22.8	-21.5	-18.8	-14.7	-6.3	-2.5		
	0.4%	WB	0.5	-1.9	-4.6	-11.2	-11.1	-7.8	-12.4	-7.0	-11.0	-7.1	-2.4	-0.5		
		MCDB	1.4	-1.5	-4.4	-11.0	-11.0	-7.3	-12.2	-7.0	-10.9	-6.9	-1.6	0.4		
	2%	WB	-1.2	-3.2	-6.8	-13.9	-13.3	-11.2	-16.5	-11.9	-13.9	-9.8	-3.9	-1.4		
		MCDB	-0.6	-2.8	-6.5	-13.7	-13.0	-10.9	-16.2	-11.7	-13.5	-9.6	-3.3	-0.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	-2.4	-4.2	-9.0	-17.1	-15.9	-17.3	-20.0	-17.2	-17.6	-12.3	-5.1	-2.1		
		MCDB	-1.9	-3.8	-8.8	-16.9	-15.6	-17.0	-19.9	-17.0	-17.4	-12.1	-4.7	-1.4		
	10%	WB	-3.5	-5.8	-12.6	-19.5	-19.5	-21.6	-23.7	-20.8	-19.9	-14.5	-6.2	-2.9		
		MCDB	-3.0	-5.4	-12.4	-19.3	-19.2	-21.4	-23.5	-20.5	-19.7	-14.2	-5.7	-2.2		
	5% DB	MDBR	6.1	8.0	8.0	8.3	8.8	8.8	9.4	9.2	9.1	8.9	7.8	6.2		
		MCDBR	6.8	7.9	10.6	11.5	11.5	12.9	15.4	13.2	12.6	9.9	7.6	6.9		
(35) Mean Daily Temperature Range	5% WB	MCWBR	5.5	7.2	9.7	11.2	9.9	12.7	15.8	12.2	13.4	9.0	7.0	5.9		
		MCDBR	5.4	7.7	10.3	11.2	9.9	12.9	15.2	12.3	12.8	9.5	7.3	6.1		
	5% WB	MCWBR	4.9	7.5	10.2	11.2	9.9	12.8	15.0	12.2	12.5	9.2	6.8	5.4		
		MCDBR	4.9	7.5	10.2	11.2	9.9	12.8	15.0	12.2	12.5	9.2	6.8	5.4		
(40) Clear-Sky Solar Irradiance	taub		0.247	0.232	0.194	0.120	N/A	N/A	N/A	0.126	0.194	0.215	0.230	0.247		
	taud		2.222	2.232	2.197	1.899	N/A	N/A	N/A	1.860	2.133	2.189	2.214	2.203		
	Ebn at Noon		961	894	773	290	0	0	0	238	747	913	980	983		
	Edh at Noon		104	83	52	10	0	0	0	10	54	86	105	113		
(44) All-Sky Solar Radiation	RadAvg		7.52	4.71	1.77	0.21	0.00	0.00	0.00	0.06	1.02	3.55	6.87	8.59		
	RadStd		0.24	0.21	0.09	0.01	0.00	0.00	0.00	0.00	0.05	0.10	0.21	0.23		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page